

# Take a Note of UI with .NET 10

**Peter Bull**

**Microsoft MVP - .NET & Windows Development**





# Peter Bull



- Senior Software Engineer for Shared Services using .NET at Klipboard
- Awarded Microsoft MVP for .NET & Windows Development
- .NET Packages for Assets & Toolkits – [comentsys.com](https://comentsys.com)
- Posts, Articles & Podcast – [rogueplanetoid.com](https://rogueplanetoid.com)
- Tutorials, Talks & Workshops – [tutorialr.com](https://tutorialr.com)
- LinkedIn – [linkedin.com/in/rogueplanetoid](https://linkedin.com/in/rogueplanetoid)
- GitHub – [github.com/rogueplanetoid](https://github.com/rogueplanetoid)

# What's New?

## **Runtime & Libraries**

.NET runtime improvements, new library APIs & support for file-based apps

## **Windows Forms & WPF**

Quality enhancements to Windows Forms and fluent style changes and improvements to WPF

## **ASP.NET Core & Blazor**

ASP.NET Core with improved form validation, Blazor reconnection UI component & state persistence

## **Extension members**

C# 14 adds syntax for extension members enable extension properties to be defined

## **Field-backed properties**

C# 14 enables property accessor body without need to declare an explicit backing field

## **Null-conditional assignment**

C# 14 has null-conditional member access operators that can be used on left side of an assignment

**dot.net**

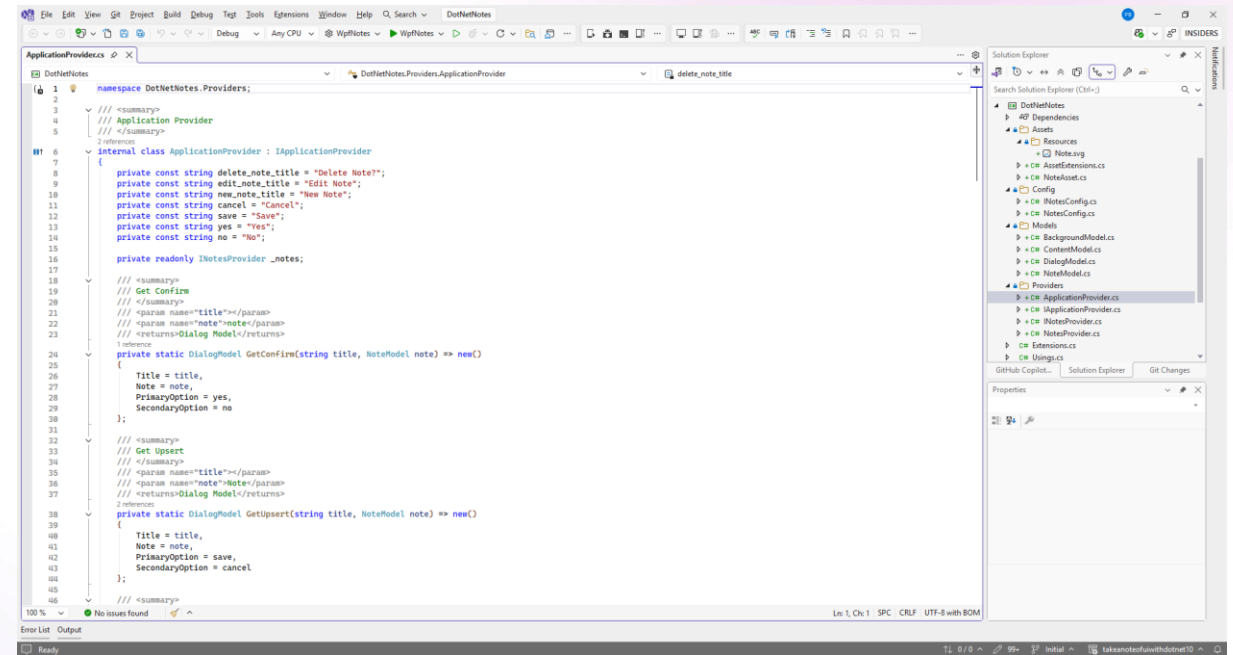
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# Visual Studio 2026



## Stay in your flow & level up with AI

- Visual Studio 2026 builds on what developers use to make it more intuitive to help developers build something new or scale existing solutions
- Visual Studio 2026 features a refreshed & modern Fluent Design with updated icons, improvements to user interface & increased performance
- Visual Studio 2026 deeply integrates with GitHub Copilot to understand your work with context across your solution, repository or tools
- Visual Studio 2026 offers compatibility for existing workloads & extensions from Visual Studio 2022 providing future-ready IDE for .NET 10



**visualstudio.com**

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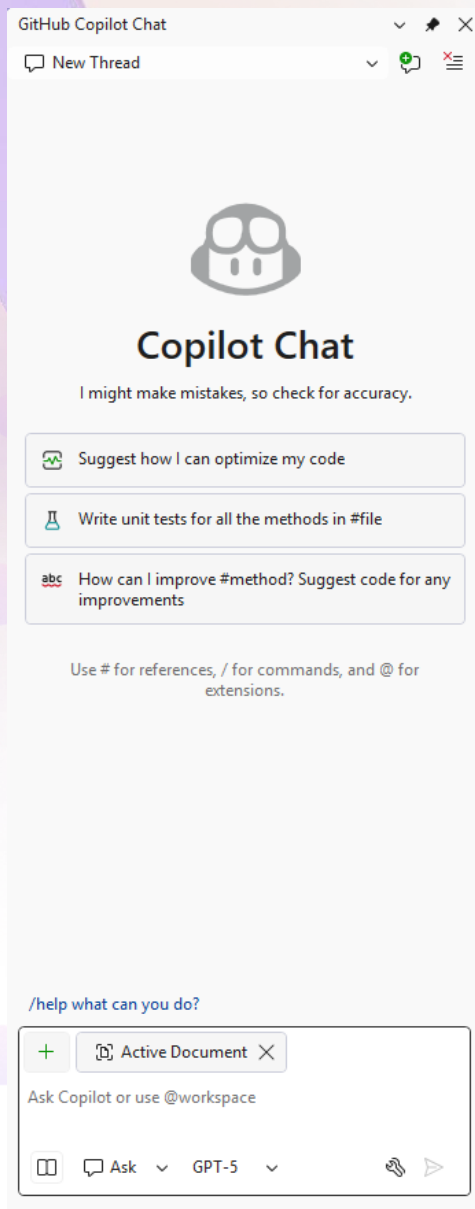
# GitHub Copilot

## AI that builds with you

- GitHub Copilot helps transform the developer experience to help focus on problem solving
- GitHub Copilot can analyse code, propose edits, create functionality, fix bugs or mistakes
- GitHub Copilot can perform actions by leveraging a wide range of tools using MCP servers
- GitHub Copilot has modes of ask to control changes or agent to apply changes

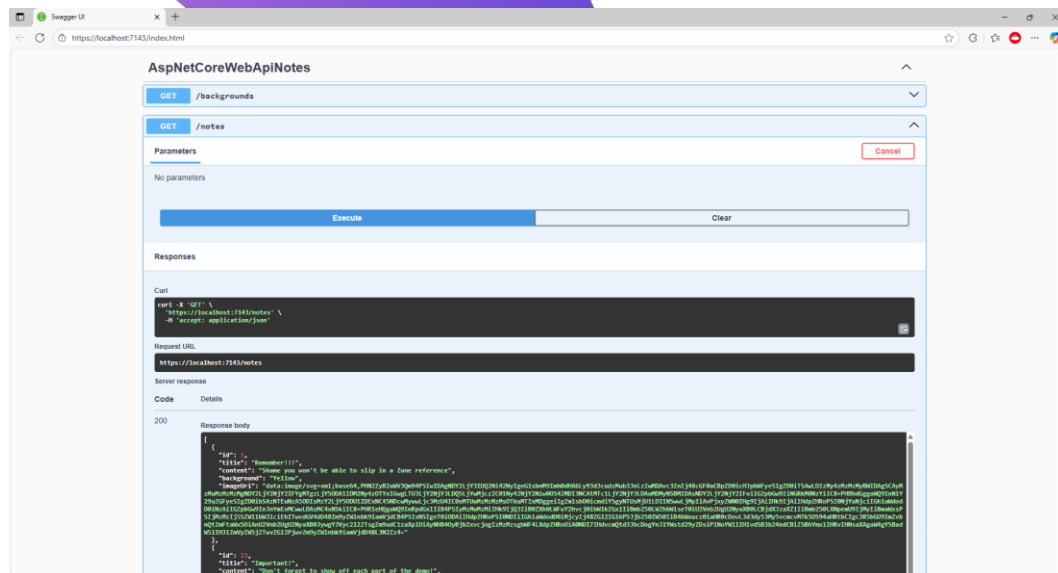
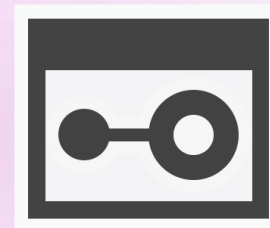
[github.com/copilot](https://github.com/copilot)

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# UI in .NET

**A to (almost) Z of first-party UI in .NET & how GitHub Copilot helped**

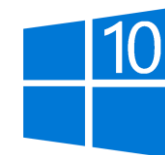


# ASP.NET Core Web API

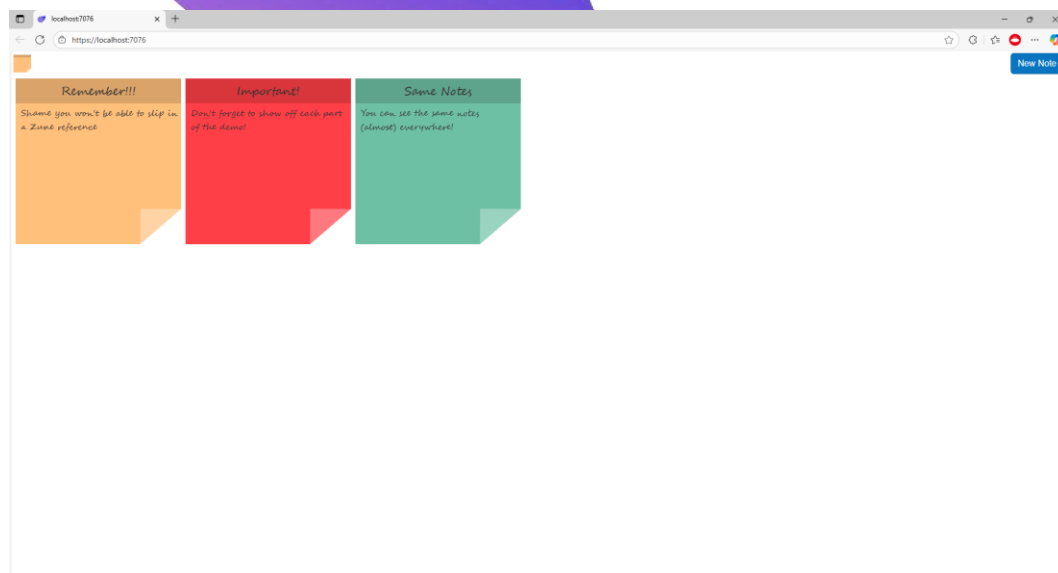
## RESTful Web APIs using ASP.NET Core controllers or minimal APIs



- Create suitable API endpoints from a Notes application
- Create an allowed background attribute
- Create a .http file for my API







# Blazor Web App

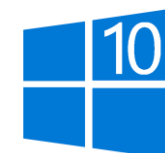
**Blazor web app supporting server-side rendering and client interactivity**

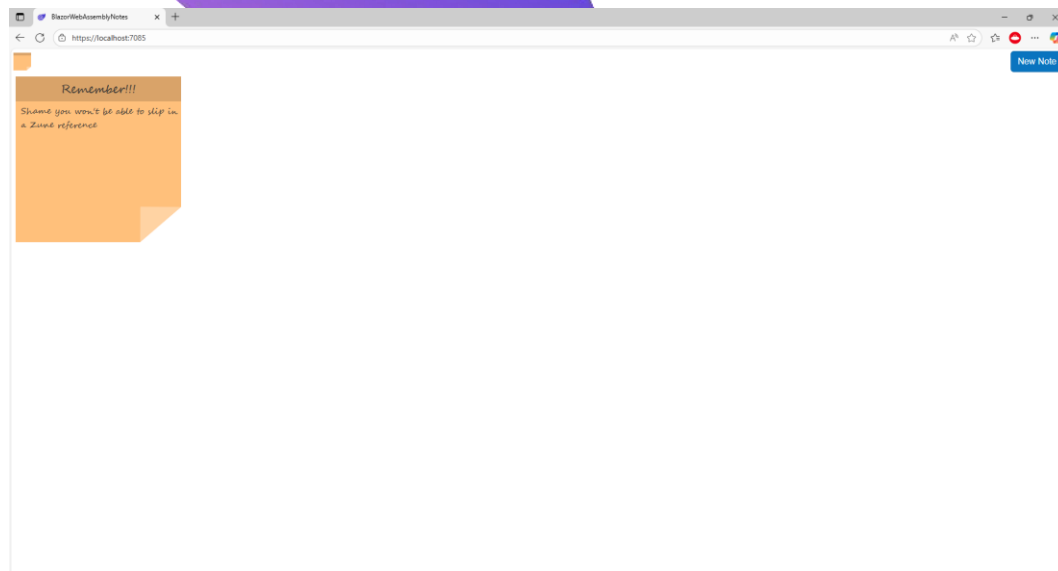


- Update modal to support edit and new



macOS





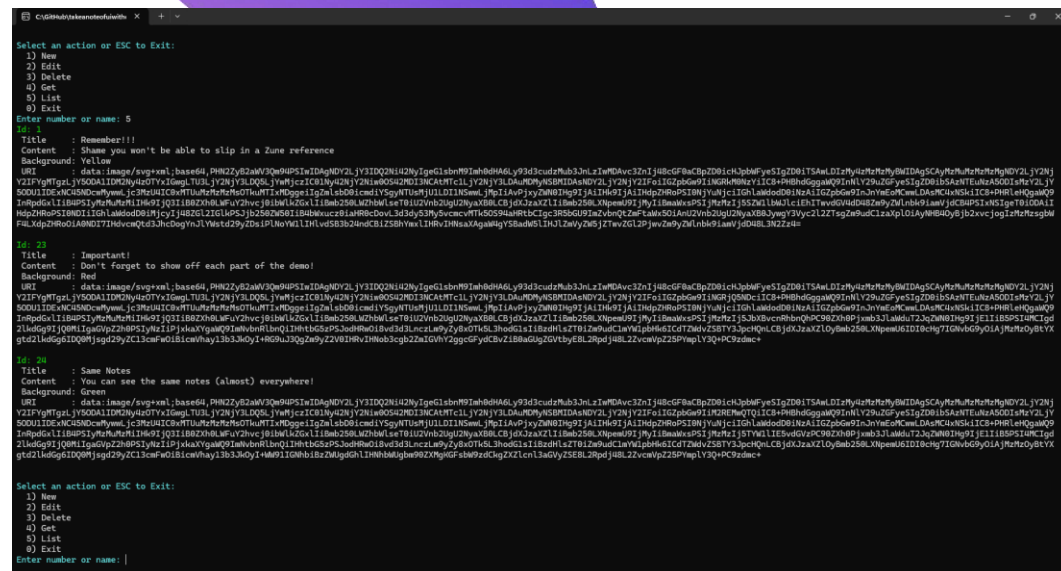
# Blazor WebAssembly App

**Blazor WebAssembly for web apps  
with rich dynamic user interfaces**



- Create local storage-based based notes provider





# Console App

## Command-line application for Linux, macOS & Windows



- Create a Console application based upon Notes application
- Create command-line argument parser
- Update with interactive mode if no arguments provided
- Update to accept valid backgrounds
- Update to quit interactive mode with Esc
- Update to confirm before deletion 🐧

  
macOS

## Take a Note of UI with .NET 10



```
DotNetNotes - NoteModel.cs
NoteModel.cs
namespace DotNetNotes.Models;

/// <summary>
/// Note Model
/// </summary>
/// </summary>
public class NoteModel : ObservableBase
{
    private int? _id = null;
    private string _title = string.Empty;
    private string _content = string.Empty;
    private string _background = "#ffc075";

    /// <summary>
    /// Id
    /// </summary>
    /// </summary>
    public int? Id
    {
        get => _id;
        set => SetProperty(ref _id, value);
    }

    /// <summary>
    /// Identifier
    /// </summary>
    /// </summary>
    public string? Identifier
    {
        get => Id?.ToString();
        set => Id = int.TryParse(value, out var id) ? id : null;
    }

    /// <summary>
    /// Title
    /// </summary>
    /// </summary>
    [Required]
    [StringLength(60)]
    public string Title
    {
        get => _title;
        set => SetProperty(ref _title, value);
    }

    /// <summary>
    /// Content
    /// </summary>
    /// </summary>
    [Required]
    [StringLength(255)]
    public string Content
    {
        get => _content;
        set => SetProperty(ref _content, value);
    }
}
```

# Class Library

Reusable collection of classes, interfaces & methods



- Update methods with required attributes



```
1 namespace MyNotes.Tools;
2
3 /// <summary>
4 /// Notes Tools
5 /// </summary>
6 /// <param name="application">Application</param>
7 internal class NotesTools(IApplicationProvider application)
8 {
9     private const char hash = 'M';
10    private const string unknown = "Unknown";
11
12    /// <summary>
13    /// List Notes
14    /// </summary>
15    /// <returns>List of Notes</returns>
16    [McpServerTool]
17    [Description("Get a list of all notes with their details")]
18    public async Task<object> ListNotes()
19    {
20        var notes = await application.ListAsync();
21        return notes.Select(note => new
22        {
23            Id = note.Id,
24            Title = note.Title,
25            Content = note.Content,
26            Background = note.Background,
27            BackgroundHash = BackgroundHelper.GetBackgroundHash(note.Background, unknown)
28        }).ToList();
29    }
30
31    /// <summary>
32    /// Get Note
33    /// </summary>
34    /// <param name="id">Note ID</param>
35    /// <returns>Note</returns>
36    [McpServerTool]
37    [Description("Get a specific note by its ID")]
38    public async Task<object> GetNote([Description("The ID of the note to retrieve")] int id)
39    {
40        var note = await application.GetAsync(id);
41        return note == null ? throw new InvalidOperationException("Note with ID {id} not found")
42        : (object)new
43        {
44            Id = note.Id,
45            Title = note.Title,
46            Content = note.Content,
47            Background = note.Background,
48        };
49    }
50 }
```

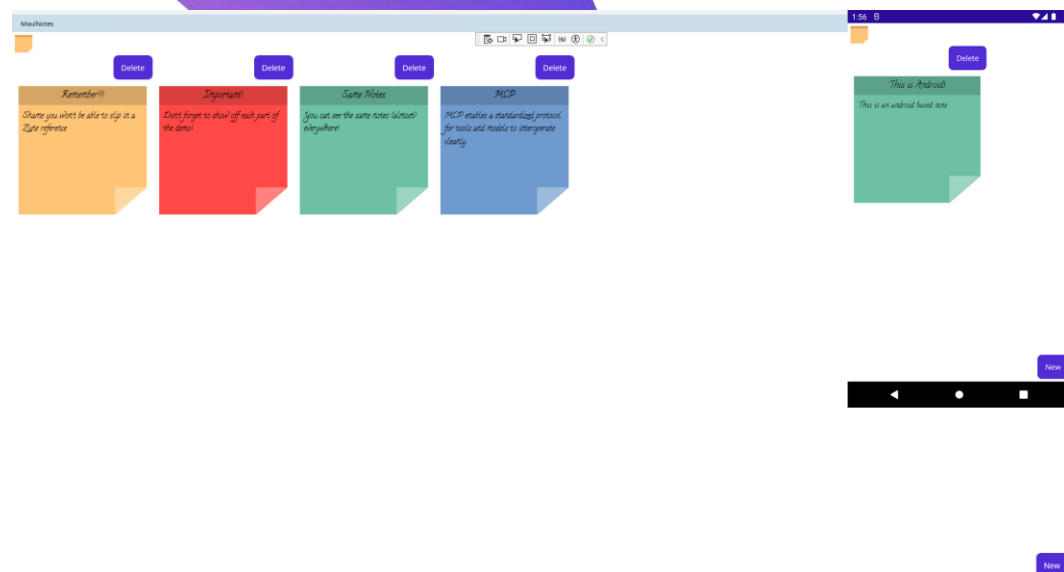
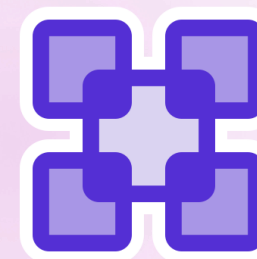
# Local MCP Server Console App

## Model Context Protocol server command-line application



- Create an MCP tool for my Notes application





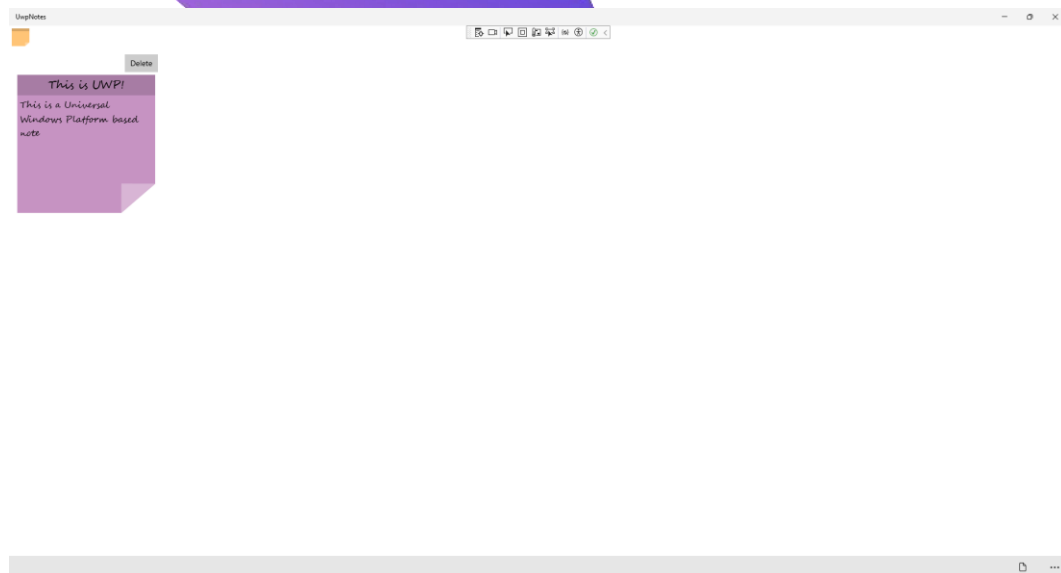
# .NET MAUI App

Application for iOS, Android, Mac Catalyst & WinUI



- Update Note Control to be scalable
- Update Popup with latest changes in .NET MAUI Community Toolkit





# UWP App

## Universal Windows Platform app using .NET & Native AOT



- Create a UWP app based upon Notes application
- Create Relay Command, View Model and Converters for Note and Content Model
- Create a HTML Colour to Windows Colour Converter
- Update Class Library methods with required attributes for AOT



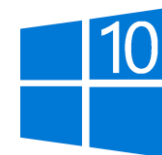


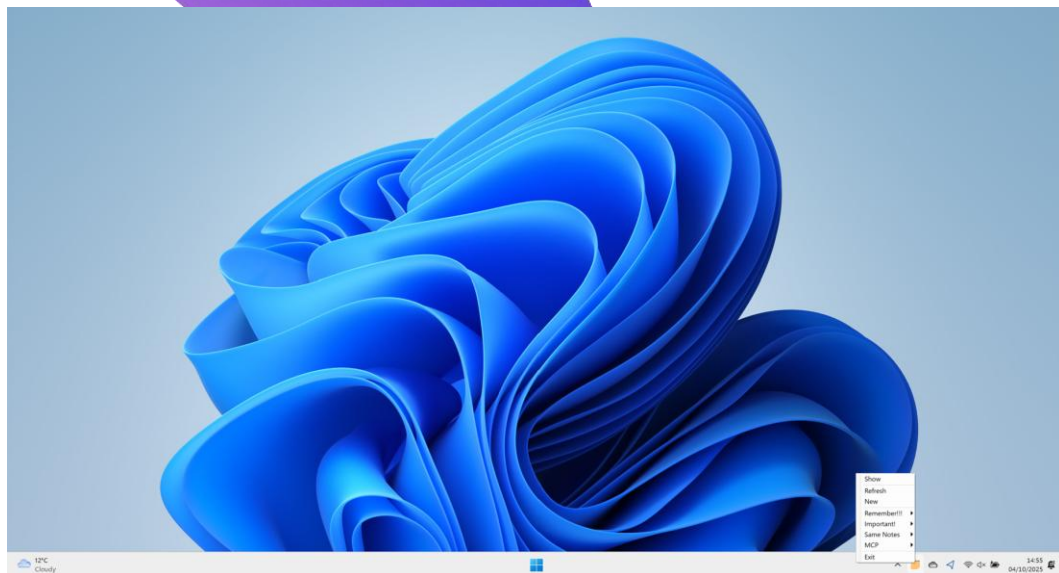
# Windows Forms App

## WinForms App



- Create a Windows Form app based upon Notes application
- Create a Note Control based upon the Note Control from Notes application
- Update Note Control so that corner is on the bottom right instead of the top right



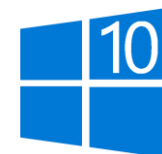


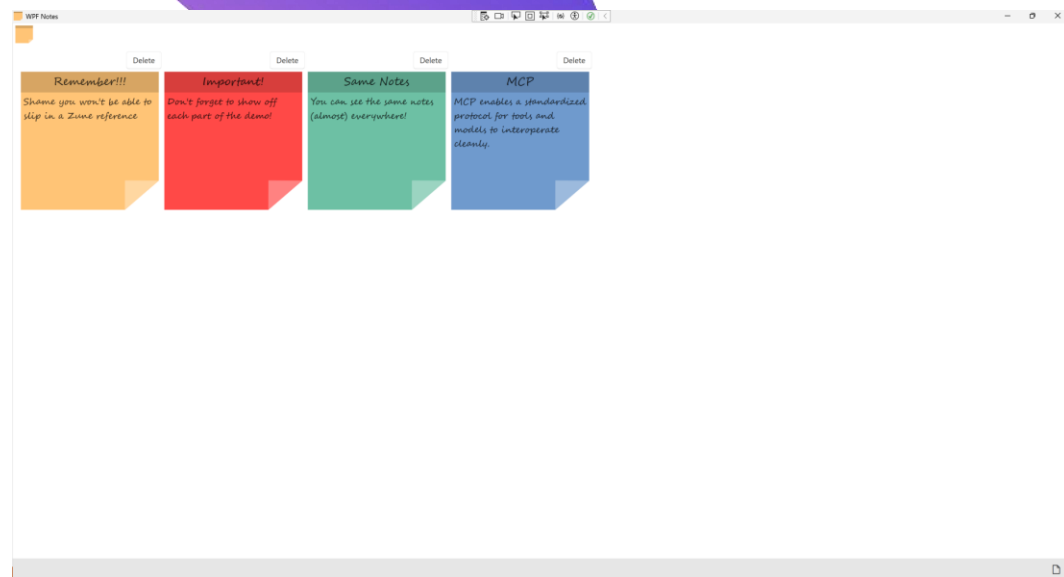
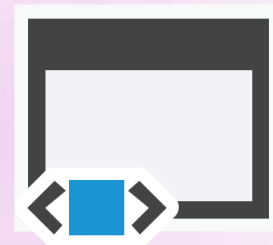
# Windows Forms Notify Icon

## Notification Area or Task Tray icon from Windows Forms App



- Create a Notification Area implementation based upon Notes application
- Update implementation with an option to exit the application
- Update implementation to be loaded when Windows Forms application is closed and hidden when Windows Forms application is loaded





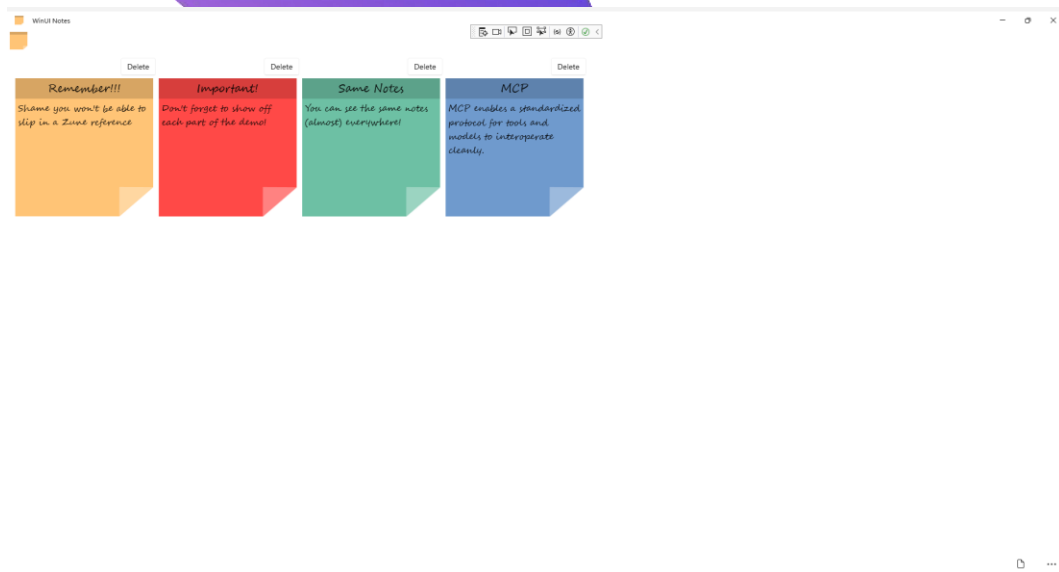
# WPF Application

## Windows Presentation Foundation app



- Create a Note Control based upon the Note Control from Notes application





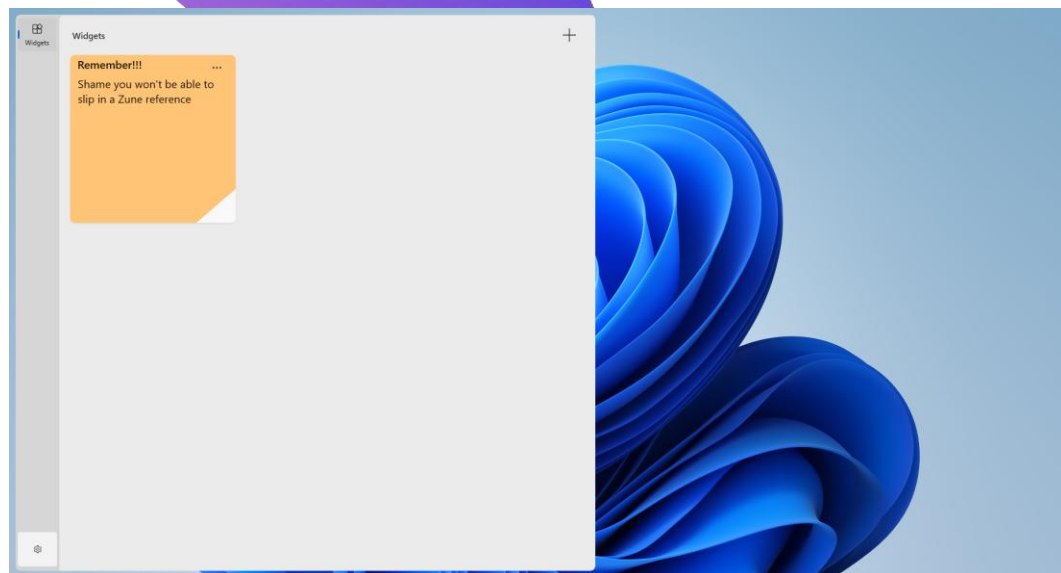
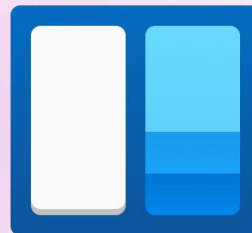
# WinUI App

## Windows App SDK & MSIX side-loading or distribution via Microsoft Store



- Create a Console App, UWP App along with Windows Form App based upon this Notes application
- Create Note Control for WPF app based on Note Control from this Notes application
- Add Titlebar from Windows App SDK 1.8





# Windows Widget

## Widget Provider from WinUI App using Windows App SDK



- Create Adaptive Cards for Configuration with New and Refresh at the top, a list of three Notes with Edit and Delete with Paging at the bottom
- Create a Confirmation Adaptive Card based on Confirmation from WinUI application
- Create an Adaptive Card for creating or editing Notes for Notes application
- Create a service for Configuration which includes support for paging of notes

# NuGet



## **Comentsys.Toolkit**

Toolkit for .NET supporting SVG-based assets & Data Binding for XAML apps



## **Comentsys.Toolkit.WindowsAppSdk**

Toolkit for Windows App SDK including support for Widget Providers in apps



## **ModernWpfUI**

Modern styles & features for Windows Presentation Foundation apps



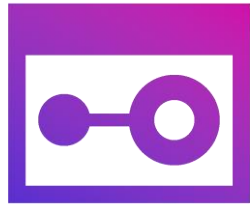
## **CommunityToolkit.Maui**

Features for .NET MAUI apps including popup for presenting information

**nuget.org**

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# .NET first-party UI technologies



**ASP.NET Core  
Web API**



**ASP.NET Core  
Web App**



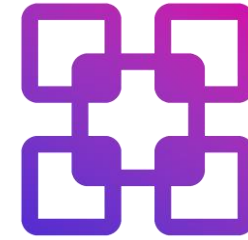
**Blazor Web &  
WebAssembly App**



**Console  
App**



**Local MCP Server  
Console App**



**.NET MAUI  
App**



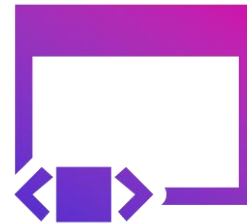
**UWP  
App**



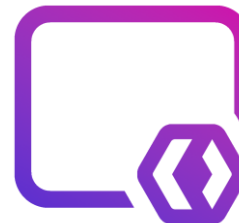
**Windows Forms  
App**



**Windows Forms  
Notify Icon**



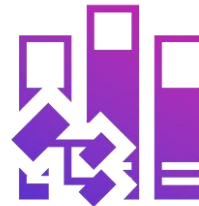
**WPF  
Application**



**WinUI  
App**



**Windows  
Widget**



**Class Library**



# Demo .NET 10 Notes Apps

**Take a Note of UI with .NET 10**

# **.NET third-party UI technologies**



**Uno Platform**



**Avalonia UI**



**OpenSilver**

# Summary



## Cloud & Web

Create cloud and web apps using ASP.NET Core Web API / Web App

Build Blazor Web apps for server or client using WebAssembly

## Console & Class Library

Craft console apps including local Model Context Protocol servers

Contain common functionality for apps with Class Libraries

## Mobile & Desktop

Make cross-platform mobile and desktop apps using .NET MAUI

Define Windows desktop apps with UWP, WinForms, WinUI, or WPF

# Questions

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Include .NET packages

**[github.com/RoguePlanetoid/takeanoteofuiwithdotnet10](https://github.com/RoguePlanetoid/takeanoteofuiwithdotnet10)**